

Khwaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - I

Course Code : CSE 0533 1101 (CSE) & PHY 0533 1101 (EEE)

Term: Spring – 24	Date: 03.01.2024	Time: 10.00 AM	Room: 327	Activity: Lecture 2
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Topics to be discussed: The Electricity

Permittivity of a medium, Coulomb's law

Target CLO & PLO : CLO1,2,4 & PLO1,2,4

Learning Outcomes : At the end of the class the students will be able to:

- ❖ State and explain Coulomb's law
- ❖ Define permittivity and relative permittivity of the medium
- ❖ Solve some recommended mathematical problems.

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10 min
Interactive discussion on objectives of the all-sub-topics	10 min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session	10 min
Others = 5 min	5min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

- 1) What is the unit of relative permittivity?
a) C b) N c) A d) F e) J
- 2) Which is possible amount of charge of an object?
a) $Q=100.8e$ b) $Q=80e$ c) $Q=13.5e$ d) $Q=9.2e$ e) $Q=4.5e$

Sample Questions (SAQ):

- 1) State and Explain Coulomb's Law.
- 2) What are the limitations of the Coulomb's Law?
- 3) Define Permittivity and Relative Permittivity.
- 4) The distance r between the electron and the proton in the hydrogen atom is about 5.3×10^{-11} m. What are the magnitudes of electric force?
- 5) Calculate the repulsive Coulomb force exist between two protons in a nucleus of iron. Assume a separation of 4×10^{-15} m.
- 6) A negative charge of 10^{-6} is situated in air at the origin of a rectangular co-ordinate system. A second negative point charge of 10^{-4} is situated on the positive X-axis at a distance of 50 cm from the origin. What is the force on the second charge?

Reference Books:

1. Giasuddin ahmed – Physics for Engineers (Part-2)
2. Huq. Rafiqullah- Electricity and Magnetism

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 326-331
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 37-86